



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 01 July 2026

In addition to Part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : ME G615
Course Title : Computer Aided Design & Additive Manufacturing
Instructor-in-Charge : Sunil Sinhmar (sunil.sinhmar@pilani.bits-pilani.ac.in, Chamber No. 2220-L)
Instructor : Murali Palla (murali.palla@pilani.bits-pilani.ac.in, Chamber No. 1215-O)

1. Course Description

This course provides a comprehensive understanding of advanced computer-aided design (CAD) and digital product development techniques used in modern engineering design and manufacturing. It covers geometric modelling of parametric curves, surfaces, and Non-Uniform Rational B-Splines (NURBS), along with geometric transformations, mass property evaluation, and product data exchange standards for seamless CAD interoperability. The course introduces Design of Experiments (DoE) for design optimisation and provides extensive hands-on training in basic and advanced Geometric Dimensioning and Tolerancing (GD&T) based on the ASME Y14.5 standard using commercial CAD software. Students will also explore the fundamentals of generative design, topology optimisation through industrial case studies, and the principles and applications of additive manufacturing, enabling them to design innovative, lightweight, and manufacturable engineering products.

2. Scope & Objective of the Course

The course aims to develop knowledge and practical skills in advanced CAD, geometric modelling, topology optimisation, and additive manufacturing techniques. It enables students to apply geometric modelling, GD&T (ASME Y14.5), generative design, and modern engineering software to design, analyse, and validate innovative and manufacturable engineering products. The course is intended to enable students to:

3. Course Learning Outcomes

At the end of the course, students will be able to:

- CLO1.** Understand the mathematical and computational foundations of parametric geometric modelling;
- CLO2.** Construct and interpret curves, surfaces, solids, assemblies, and product-data representations;
- CLO3.** Apply ASME Y14.5 concepts to communicate functional dimensional and geometric requirements;
- CLO4.** Formulate and interpret generative-design and topology-optimization problems;

CLO5. Understand additive-manufacturing processes, materials, preprocessing, process parameters, post-processing, and safety; and

CLO6. Integrate CAD, GD&T, optimization, and additive manufacturing in a coherent digital product-development workflow.

4. Textbooks

M1: Geometric Modelling

T1. Zeid, I., *Mastering CAD/CAM*, Tata McGraw-Hill, New Delhi.

T2. Mortenson, Michael E., *Geometric Modeling*, John Wiley & Sons, Inc.

M2: Geometric Dimensioning and Tolerancing GD&T

T3. Cogorno, G. R., *Geometric Dimensioning and Tolerancing for Mechanical Design*, McGraw-Hill.

M3: Generative Design

<https://www.autodesk.com/in/solutions/generative-design/manufacturing>

M4: Topology Optimization

T4. Bendsoe, Martin P. and Ole Sigmund, *Topology Optimization: Theory, Methods, and Applications*, Springer Nature.

M5: Additive Manufacturing

T5. Paul, C. P. and A. N. Jinoop, *Additive Manufacturing*, McGraw Hill.

References

R1. Mirzendehtdel, Amir M. and Krishnan Suresh, *A Hands-On Introduction to Topology Optimization*, CreateSpace Independent Publishing.

5. Course Plan

Weekly plan for 14 weeks, normally with three lecture sessions per week.

Week No.	Module No.	Description	Reference	Learning Outcomes
1	M1	L1: Introduction to CAD and digital product development L2: Homogeneous coordinates, Affine transformations, Projections L3: Parametric representation of curves	T1, T2	CLO1, CLO2
2	M1	L4: Synthetic curves and continuity; L5: Bézier curves; L6: B-spline curves	T1, T2	CLO2

Week No.	Module No.	Description	Reference	Learning Outcomes
3	M1	L7: NURBS curves and conic representation; L8: Surface modelling; L9: Solid modelling and topology	T1, T2	CLO2, CLO3
4	M1–M2	L10: Geometric transformations, assemblies, and mass properties; L11: Product-data exchange and design of experiments; L12: Need for tolerancing and foundations of GD&T	T1–T3, Class notes	CLO1–CLO3
5	M2	L13: Size tolerances, envelope principle, MMC, LMC, and RFS; L14: Datum fundamentals and degrees of freedom; L15: Advanced datum systems	T3	CLO3
6	M2	L16: Form tolerances; L17: Orientation tolerances; L18: Profile tolerances	T3	CLO3
7	M2	L19: Position tolerancing; L20: Runout tolerances and rotational components; L21: Virtual condition, resultant condition, and special calculations; L22: Composite tolerances, model-based definition, and case study;	T3	CLO3
8	M2–M4	L23: Introduction to generative design; L24: Introduction to topology optimisation	T4, Class notes	CLO3, CLO4
9	M4	L25: Principles, fundamentals and applications of topology optimisation–I; L26: Principles, fundamentals and applications of topology optimisation–II; L27: Differences of topology optimisation from traditional design processes	T4	CLO4
10	M4–M5	L28: Case studies in product and automotive sectors; L29: SIMP algorithm; L30: Introduction to additive manufacturing	T4, T5, Class notes	CLO4, CLO5
11	M5	L31: Additive manufacturing data formats and pre-processing; L32: Additive manufacturing methods–I; L33: Additive manufacturing methods–II	T5	CLO5
12	M5	L34: Design for additive manufacturing; L35: Additive manufacturing equipment and sub-systems; L36: Additive manufacturing materials	T5	CLO5
13	M5	L37: Additive manufacturing post-processing; L38: Safety considerations in additive manufacturing; L39: Applications of additive manufacturing	T5	CLO5
14	M5	L40: Cost and time saving case studies	Class notes	CLO5

6. Evaluation Scheme

Component	Wt. (%)	Duration	Date & Time	Comments
Mid-Semester Test	25	90 min	05/10/26	Closed Book
Projects/Seminars	30	–	TBA	Open Book
Quiz/Viva/Assignment	10	–	TBA	Closed/Open Book
Comprehensive Examination	35	180 min	11/12/26	Closed Book

NC Criterion / Minimum Passing Marks: The passing marks for this course will be determined based on one of the following two criteria: 30% of the median of the post-comprehensive examination score, or 20% of the average score of the first three top scorers. The lower of these two values will be considered the passing mark. NC will be awarded if a student does not appear in the mid-semester or comprehensive examination.

7. Attendance Policy

Attendance requirements will be governed by the applicable Institute rules and course instructions announced by the Instructor-in-Charge.

8. Office Consultation Hour

Monday, 12:00–01:00 p.m., Chamber 2220-L, FD2. Students may also seek a prior appointment through email.

9. Notices

All notices regarding the course will be sent through email and/or Nalanda.

10. Make-up Policy

Make-up will be granted **only in genuine cases** with prior permission, as per Institute rules. No make-up will be allowed for the Project, Quiz, Seminar, or Evaluative Tests.

Sunil Sinhmar
Instructor-in-Charge
ME G615